

CS & IT ENGINEERING

Programming in C

Control Flow Statements

Lec- 05 More about loops



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A stylized illustration of a laptop with a blue frame and an orange base. The screen is white and contains the text 'TOPICS TO BE COVERED' in a dark blue, hand-drawn font.

TOPICS TO BE
COVERED

A dotted orange arrow originates from the right side of the laptop screen, moves horizontally to the right, then turns 90 degrees downward, and finally turns 90 degrees to the right again, pointing into the left side of the blue box.

More on Loops

$$1 + 2 + 3 + 4$$

SumTillNow = 0;

SumTillNow = SumTillNow + 1

SumTillNow = SumTillNow + 2

SumTillNow = SumTillNow + 3

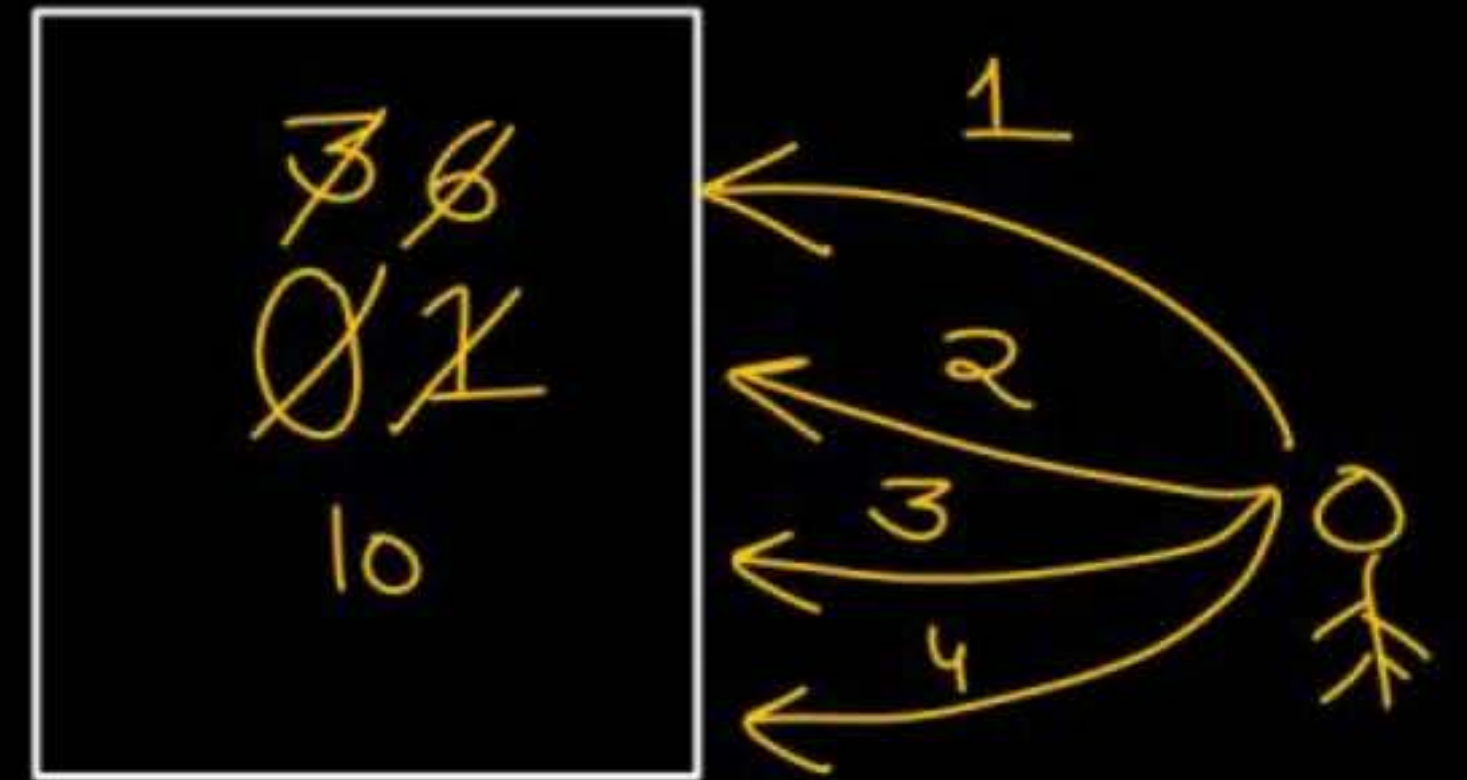
SumTillNow = SumTillNow + 4

SumTillNow = 0;

for (i = 1; i <= 4; i++)

SumTillNow = SumTillNow + i;

SumTillNow




```
SumTillNow = 0;  
for(i=1; i<=104; i++)  
    SumTillNow = SumTillNow + i;  
printf("/d", sumTillNow);
```

```
Sum = 0;  
→ for(i=1; i<=n; i++)  
    Sum = Sum + i;  
printf("/d", sum);
```

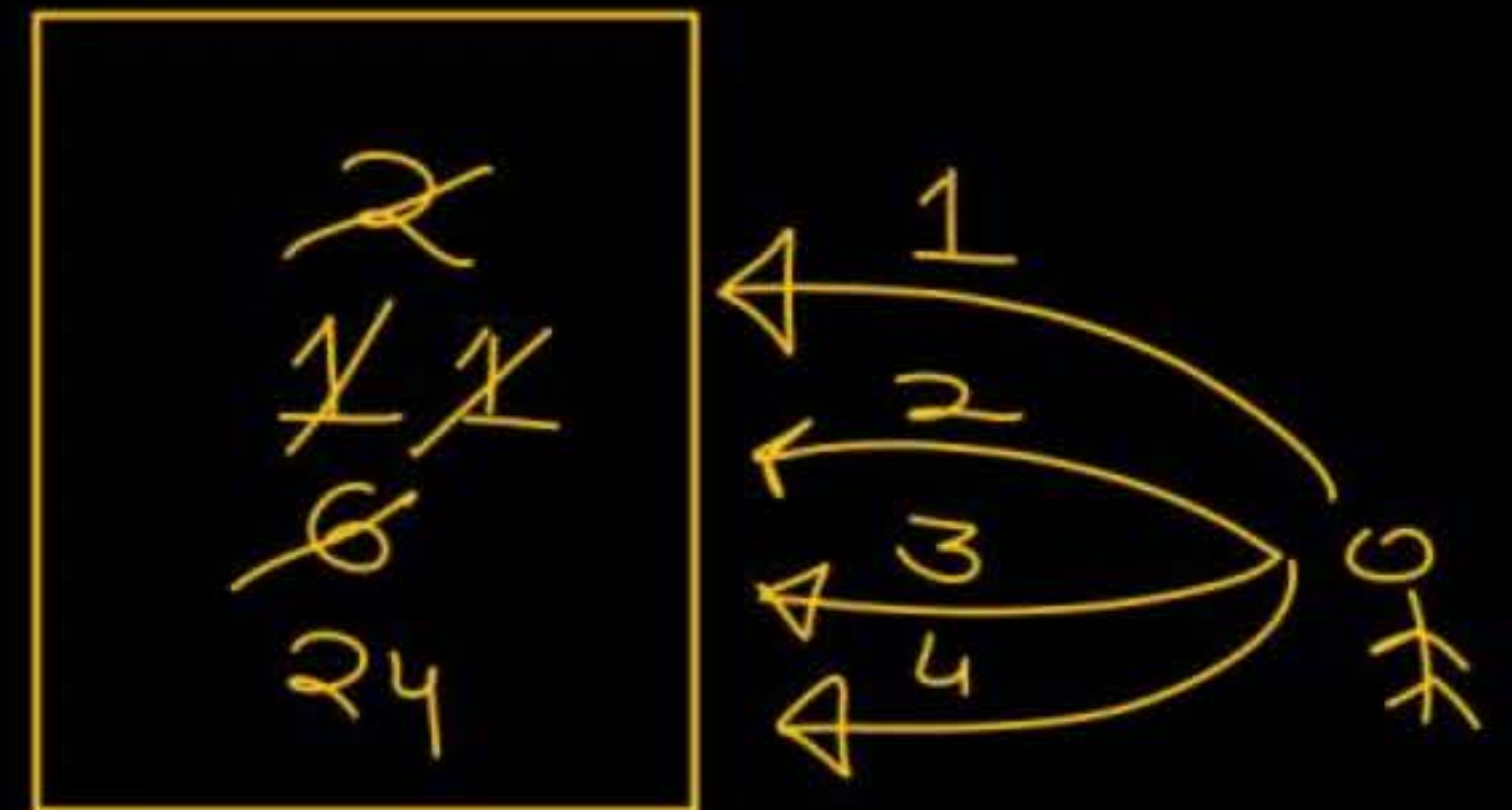
$$1 \times 2 \times 3 \times 4$$

$$\text{Prod} = 1 ;$$

$$\begin{aligned} \text{Prod} &= \text{Prod} \times 1 ; \\ \text{Prod} &= \text{Prod} \times 2 ; \\ \text{Prod} &= \text{Prod} \times 3 ; \\ \text{Prod} &= \text{Prod} \times 4 ; \end{aligned}$$

$\text{Prod} = 1$
for($i = 1; i \leq 4; i++$)
 $\text{Prod} = \text{Prod} \times i ;$

Prod



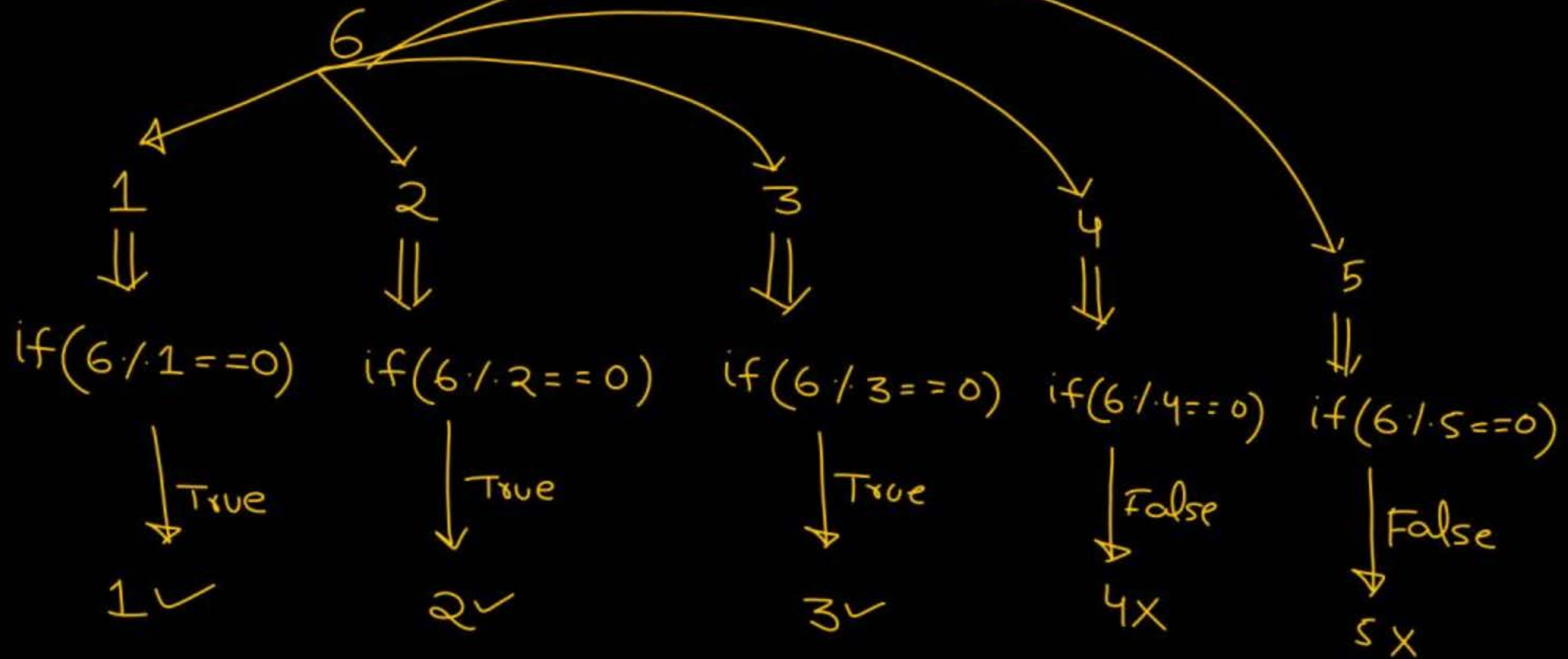
```
Prod = 1;  
for (i = 1; i <= 104; i++)  
    Prod = Prod * i;
```

```
Prod = 1  
for (i = 1; k = n; i++)  
    Prod = Prod * i;  
printf("%d", Prod);
```


12 $\rightarrow$ 2 is a factor of 12 or not?

if $(12 / 2 == 0)$

$\Rightarrow$ 2 is a factor of 12



```
for(i=1; i<6; i++)  
{  
    if(6 / i == 0)  
    {  
        i is a factor of n  
    }  
}
```



```

sum = 0;
for (i = 1; i < n; i++)
{
    if (n % i == 0)
    {
        Sum = sum + i;
    }
}

```



```

    }
    if (sum == n)
        printf("%d is a perfect number", n);
    else
        printf("%d is not a perfect number", n);
}

```

$n > 0$

I/P : 125

O/P : 3

I/P : $n = 1238$

O/P : 4

I/P : 3

O/P : 1

$n = 53$

$\downarrow n = n/10 \checkmark$
 $n = 5$

$\downarrow n = n/10 \checkmark$
 $n = 0$

$n = 1258$

$\downarrow n = n/10 \checkmark$

$n = 125$

$\downarrow n = n/10 \checkmark$
 $n = 12$

$\downarrow n = n/10 \checkmark$
 $n = 1$

$\downarrow n = n/10 \checkmark$
 $n = 0$

$1258/10$

$n \leftarrow 125$

$\swarrow 125/10$
 $n \leftarrow 12$

While($n \neq 0$)

{

$n = n/10;$

}

$n > 0$

$count = 0;$

$0! = 0$

while($n! = 0$)

{ $count++;$

$n = n/10;$

}



$n = 123$

① $123! = 0 \rightarrow \text{True}$

$n = n/10 \checkmark$

② $12! = 0 \rightarrow \text{True}$

$n = n/10 \checkmark$

③ $1! = 0 \rightarrow \text{True}$

$n = n/10 \checkmark$

$0! = 0 \Rightarrow$

$\Rightarrow$ +ve No's

while($n > 0$)

{

}

Armstrong Number

Easy version

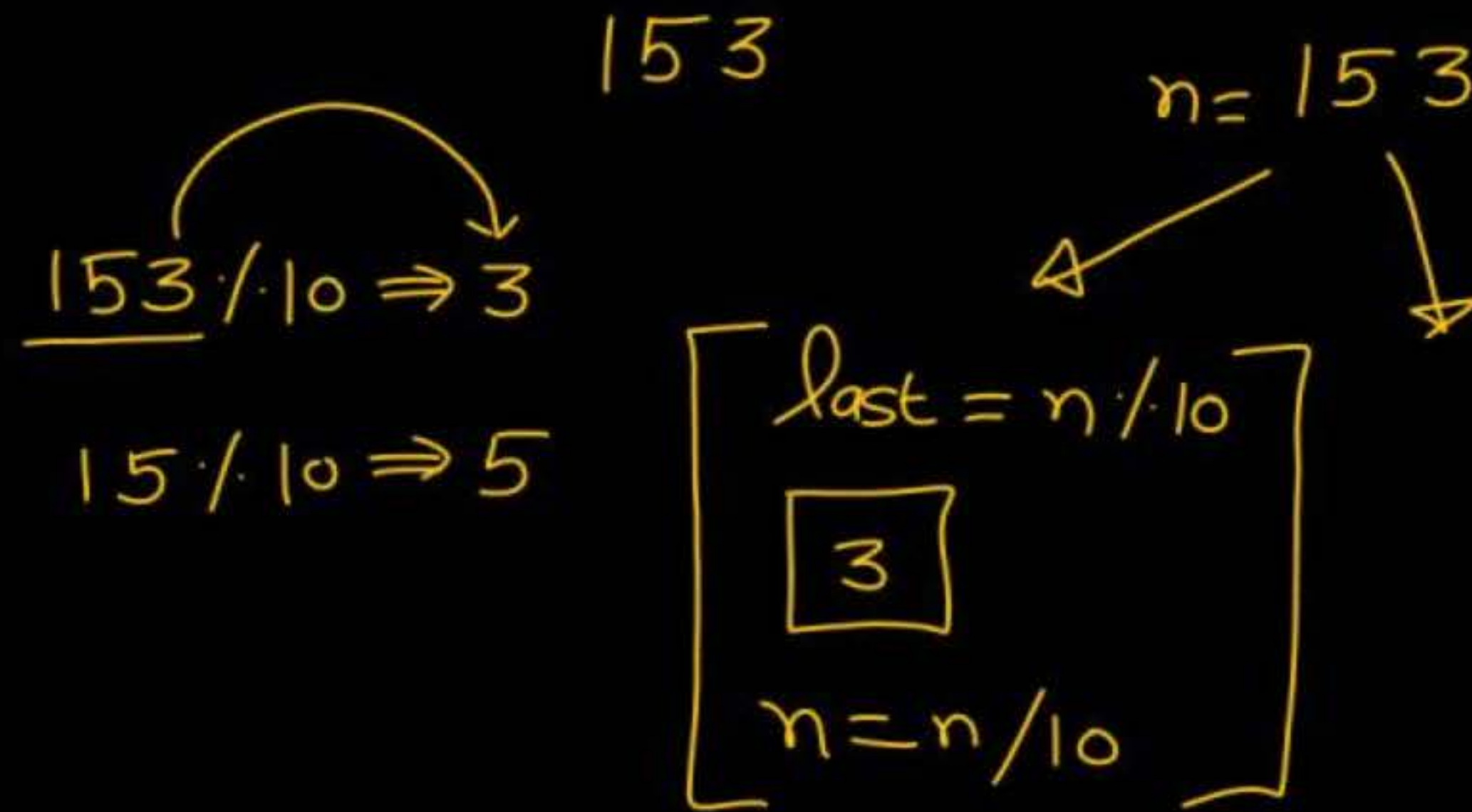
correct version

153

$$1^3 + 5^3 + 3^3$$

$$1 + 125 + 27$$

153



① $n = 153$

$\left[\begin{array}{l} \text{last} = n / 10; \boxed{3} \\ n = n / 10; \boxed{15} \end{array} \right]$

n

②

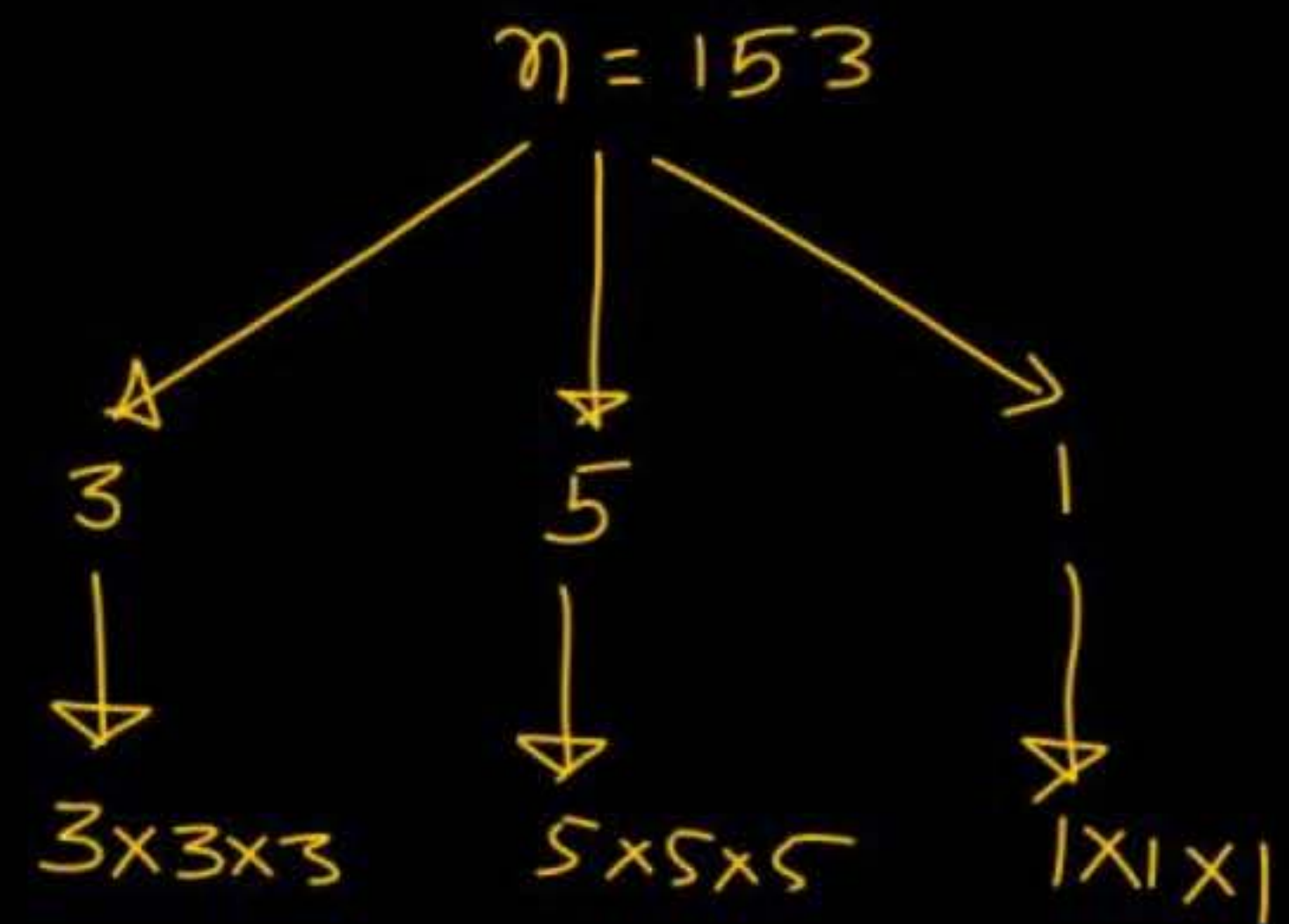
$\left[\begin{array}{l} \text{last} = n / 10 \\ n = n / 10 \end{array} \right]$

$\boxed{5}$
 last

③

$\left[\begin{array}{l} \text{last} = 1 / 10 = \boxed{1} \\ n = n / 10 = \boxed{0} \end{array} \right]$

$\boxed{1}$
 n



sum = sum + ~

153
 ↓
 15
 ↓
 1

$n = 153$: $a = n$;

$sum = 0$;

While ($n \neq 0$)

{

① $last = n / 10$;

② $n = n / 10$;

③ $Sum = sum + last \times last \times last$;

}

if ($sum == a$)

printf(——);

① $153 \neq 0 \checkmark$ $last = 3$

$n = 15$

② $15 \neq 0 \checkmark$ $last = 5$

$n = 1$

③ $1 \neq 0 \checkmark$ $last = 1$

$n = 0$

④ $0 \neq 0 \rightarrow \times$

1634

$$1^4 + 6^4 + 3^4 + 4^4$$

$$1 + 1296 + 81 + 256$$

1634

$$\sum d^{\text{No. of digits}} = N$$

① find the no. of digits

```
count = 0
While (n != 0)
{
    count ++ ;
    n = n/10;
}
```

count = 4


```
count = 0;  
while(n != 0)  
{  
    count++;  
    n = n/10;  
}
```

$1 \times 2 \times 3 \times 4 \times 5$

```
Prod = 1  
for(i = 1; i <= 5; i++)  
    Prod = Prod * i
```

$4 \times 4 \times 4 \times 4 \times 4$

```
Prod = 1  
for(i = 1; i <= 5; i++)  
    Prod = Prod * 4;
```

4^5

$a \times a \times \dots$ b times

a^b $\left[\begin{array}{l} \text{Prod} = 1 \\ \text{for}(i = 1; i \leq b; i++) \\ \text{Prod} = \text{Prod} \times a; \end{array} \right.$

$\leftarrow a^b$

$b = a = n;$

```
Count = 0;  
if (n == 0) count = 1;  
while (n != 0) {  
    count++;  
    n = n/10;  
}
```

Count $\Rightarrow$ No. of digits
 $= 4$

$a = 1634$

sum = 0

```
while (a != 0)  
{
```

last = a / 10;

a = a / 10;
prod = 1;

```
for (i = 1; i <= count; i++)  
    prod = prod * last;
```

sum = sum + prod;

```
}  
if (sum == b)
```

else — NOT

1634

$$1^4 = 1$$

$$6^4 = 1296$$

$$3^4 = 81$$

$$4^4 = 256$$

21
1297

81

256

1634

H.W

Single digit sum

without
using
any loop

i/p : 888

o/p : 6

$$8+8+8 = 24$$

$$= 2+4$$

$$= 6 \checkmark$$

i/p : 55539

o/p : 9

$$5+5+5+3+9$$

$$= 27$$

$$= 2+7$$

$$= 9$$

